Clearly, the constant ξ is infeasible in practice, however, the applied test statistic has a self normalizing property, so in the limit this constant will disappear. Now, Theorem 1.2 leads to the test-statistic

$$\widehat{W}_{m,n} = \frac{\widehat{D}^2_{m,n} - \Delta}{\widehat{V}_{m,n}} \tag{1.8}$$

and suggests to reject the hypothesis H_0 in (1.7) if $\widehat{W}_{m,n} > q_{1-\alpha}$, where $q_{1-\alpha}$ is the $(1-\alpha)$-quantile of

$$W = \frac{B(1)}{\left\{\int_0^1 s^2\big(B(s) - sB(1)\big)^2 \,\nu(\mathrm{d}s)\right\}^{1/2}}. \tag{1.9}$$

As an immediate consequence we have that the test (1.8) is asymptotically consistent for the relevant hypotheses in (1.7):

Theorem 1.3. *Let the conditions of Theorem 1.2 be satisfied. Then*

$$\lim_{m,n\to\infty} \mathbb{P}(\widehat{W}_{m,n} > q_{1-\alpha}) = \begin{cases} 1 & , D^2 > \Delta, \\ \alpha & , D^2 = \Delta, \\ 0 & , D^2 < \Delta. \end{cases} \tag{1.10}$$

Proof of Theorem 1.3. The proof is immediate. We have

$$\mathbb{P}(\widehat{W}_{m,n} > q_{1-\alpha}) = \mathbb{P}\left(\frac{\widehat{D}^2_{m,n} - D^2}{\widehat{V}_{m,n}} > q_{1-\alpha} + \frac{\Delta - D^2}{\widehat{V}_{m,n}}\right)$$

and Theorem 1.2 implies $(\widehat{D}^2_{m,n} - D^2)/\widehat{V}_{m,n} \Rightarrow W$ as well as

$$\frac{\Delta - D^2}{\widehat{V}_{m,n}} \to \begin{cases} -\infty & , D^2 > \Delta, \\ 0 & , D^2 = \Delta, \\ \infty & , D^2 < \Delta. \end{cases}$$

This proves the assertion. □

1.2. A test statistic for relevant differences of inco-variances

We derive a test for the second problem (1.7), that is whether the difference $D = \sigma^2_{\mathcal{X}} - \sigma^2_{\mathcal{Y}}$ of the independent copy variances is substantial. To this end, we consider the empirical counterpart $\widehat{D}_{m,n} := \widehat{D}_{m,n}(1,1)$, where

$$\widehat{D}_{m,n}(s,t) := \widehat{\sigma}^2_{\mathcal{X}}(s,t) - \widehat{\sigma}^2_{\mathcal{Y}}(s,t), \qquad 0 \le s,t \le 1, \tag{1.11}$$

is the difference of the two parameter inco-variance partial sum processes

$$\begin{aligned}\widehat{\sigma}^2_{\mathcal{X}}(s,t) &= \frac{1}{m(m-1)}\sum_{i=1}^{\lfloor ms\rfloor}\sum_{\substack{j=1\\ j\neq i}}^{\lfloor mt\rfloor}\frac{1}{2}W_r^2(\mathrm{PD}(\mathcal{X}_i),\mathrm{PD}(\mathcal{X}_j)),\\ \widehat{\sigma}^2_{\mathcal{Y}}(s,t) &= \frac{1}{n(n-1)}\sum_{i=1}^{\lfloor ns\rfloor}\sum_{\substack{j=1\\ j\neq i}}^{\lfloor nt\rfloor}\frac{1}{2}W_r^2(\mathrm{PD}(\mathcal{Y}_i),\mathrm{PD}(\mathcal{Y}_j)).\end{aligned} \tag{1.12}$$

We intentionally reuse some of the symbols of the previous section here to keep the notation simple and since it is always clear from the context, which testing problem is considered. Also we put $\widehat{\sigma}^2_{\mathcal{X}} = \widehat{\sigma}^2_{\mathcal{X}}(1,1)$ and $\widehat{\sigma}^2_{\mathcal{Y}} = \widehat{\sigma}^2_{\mathcal{Y}}(1,1)$, so that $\widehat{D}_{m,n} = \widehat{\sigma}^2_{\mathcal{X}} - \widehat{\sigma}^2_{\mathcal{Y}}$. Moreover, in order to exploit the results of Section 2, we introduce the kernel function

$$h\colon \mathrm{N}\times\mathrm{N}\to\mathbb{R}, \qquad h(\cdot,\cdot) := 2^{-1}W_r^2(\mathrm{PD}(\cdot),\mathrm{PD}(\cdot)), \tag{1.13}$$

so that (1.12) takes the form of U-statistics with kernel h.

Then, given a tolerance $\Delta > 0$, the null hypothesis is rejected if $\widehat{D}^2_{m,n} > \Delta$. Again, the main difficulty is to make the test statistic $\widehat{D}^2_{m,n}$ pivotal. To that end, let us introduce the following self-normalizer

$$\widehat{V}_{m,n} := \left\{\int_0^1\int_0^1 \left[\widehat{D}^2_{m,n}(s,t) - (st\widehat{D}_{m,n})^2\right]^2 \nu(\mathrm{d}s,\mathrm{d}t)\right\}^{1/2}, \tag{1.14}$$

where ν is a probability measure on $[0,1]^2$.

Theorem 1.4. *Suppose* $\lim_{n,m\to\infty} m/(m+n) = \tau \in (0,1)$. *Either assume*

Case (a): $\#\mathcal{X}_i, \#\mathcal{Y}_j \le C$ *with probability 1 for all* $i,j\in\mathbb{Z}$ *for a certain* $C\in\mathbb{R}_+$ *and*

$$\sum_{m\ge1} m\ \mathbb{E}[d_H(\mathcal{X}_0,\mathcal{X}_0^{(m)})^{1-1/r}] < \infty \text{ and } \sum_{m\ge1} m\ \mathbb{E}[d_H(\mathcal{Y}_0,\mathcal{Y}_0^{(m)})^{1-1/r}] < \infty$$

or assume

Case (b): Both $d<r$ *as well as*

$$\sum_{m\ge1} m\ \mathbb{E}[d_H(\mathcal{X}_0,\mathcal{X}_0^{(m)})^{\rho}] < \infty \text{ and } \sum_{m\ge1} m\ \mathbb{E}[d_H(\mathcal{Y}_0,\mathcal{Y}_0^{(m)})^{\rho}] < \infty$$

for some $\rho\in(0,1-d/r)$.

Then

$$\begin{aligned}&\sqrt{m+n}(\widehat{D}^2_{m,n} - D^2, \widehat{V}_{m,n})\\ &\Rightarrow \left(2\xi B(1), \left\{\xi^2\int_0^1\int_0^1 \left[st\big(tB(s)+sB(t)-2stB(1)\big)\right]^2 d\nu(s,t)\right\}^{1/2}\right),\end{aligned} \tag{1.15}$$

where $\xi = 2\sqrt{\frac{\Gamma_{\mathcal{X}}}{\tau} + \frac{\Gamma_{\mathcal{Y}}}{1-\tau}}(\sigma_{\mathcal{X}}^2 - \sigma_{\mathcal{Y}}^2)$ *and where* $\Gamma_{\mathcal{X}} = \sum_k \operatorname{Cov}(h_{\mathcal{X}}(\mathcal{X}_0), h_{\mathcal{X}}(\mathcal{X}_k))$, *resp.,* $\Gamma_{\mathcal{Y}} = \sum_k \operatorname{Cov}(h_{\mathcal{Y}}(\mathcal{Y}_0), h_{\mathcal{Y}}(\mathcal{Y}_k))$ *for*

$$
\begin{aligned}
h_{\mathcal{X}}(\cdot) =& \mathbb{E}\Big[\frac{1}{2} W_r^2(\mathrm{PD}(\cdot), \mathrm{PD}(\mathcal{X}'))\Big], \qquad \mathcal{X}' \sim \mathbb{P}_{\mathcal{X}},\\
h_{\mathcal{Y}}(\cdot) =& \mathbb{E}\Big[\frac{1}{2} W_r^2(\mathrm{PD}(\cdot), \mathrm{PD}(\mathcal{Y}'))\Big], \qquad \mathcal{Y}' \sim \mathbb{P}_{\mathcal{Y}}.
\end{aligned}
$$

Theorem 1.4 leads to the test-statistic

$$
\widehat{W}_{m,n} = \frac{\widehat{D}_{m,n}^2 - \Delta}{\widehat{V}_{m,n}}.
$$

and suggests to reject the hypothesis H_0 in (1.7) if $\widehat{W}_{m,n} > q_{1-\alpha}$, where $q_{1-\alpha}$ is the $(1-\alpha)$-quantile of

$$
W = \frac{2B(1)}{\Big\{\int_0^1 \int_0^1 \Big[st\big(tB(s) + sB(t) - 2stB(1)\big)\Big]^2 d\nu(s,t)\Big\}^{1/2}}. \tag{1.16}
$$

As immediate consequence we have that the test is asymptotically consistent for the relevant hypotheses in (1.7). The proof works in the same fashion as the proof of Theorem 1.3 and we skip it.

Theorem 1.5. *Let the regularity conditions of Theorem 1.4 be satisfied. Then*

$$
\lim_{m,n\to\infty} \mathbb{P}(\widehat{W}_{m,n} > q_{1-\alpha}) = \begin{cases} 1 & , D^2 > \Delta,\\ \alpha & , D^2 = \Delta,\\ 0 & , D^2 < \Delta. \end{cases}
$$

2. Mathematical background: a two-parameter FCLT for U-statistics

In this section we derive a general functional central limit theorem for (bivariate) U-statistics: Let (M,d) denote a metric space and let $h\colon M \times M \to \mathbb{R}$ be a symmetric and measurable kernel.

Let $(X_t)_{t\in\mathbb{Z}}$ be a stationary sequence of M-valued random elements defined on some common probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with marginal distribution $X_t \sim \mathbb{P}_X$, $t \in \mathbb{Z}$. The corresponding U-statistic is defined by

$$
U_n(h) = \frac{1}{n(n-1)} \sum_{1\le i\le n} \sum_{\substack{1\le j\le n\\ j\ne i}} h(X_i, X_j) = \frac{2}{n(n-1)} \sum_{1\le i<j\le n} h(X_i, X_j). \tag{2.1}
$$

Suppose X' is an independent copy of X_1, i.e., X' is independent of X_1 and has the same law $\mathbb{P}_X$. Then (2.1) is a possible estimator for $\theta := \mathbb{E}[h(X_1, X')]$. It is well known that there is a close relation between the U-statistic $U_n(h)$ and UMV-estimators for θ in the special case of an underlying i.i.d. sequence $(X_t)_t$